



IDTP9030/9020: IDT introduced wireless power transmitter / receiver solution

Some of the firms providing wireless power chipsets

- Texas Instruments
- IDT
- Qualcomm
- Toshiba
- Freescale Semiconductor
- ON Semiconductor
- NXP Semiconductor
- Panasonic

older phone models. Most of these wireless charging products are also smart to a certain extent—for example, they can automatically switch off power transfer once the device is fully charged.

Chip industry leaders, including TI, Freescale, IDT and Intel, are getting deeply involved in wireless power, enabling OEMs to easily fit in such capabilities into their products. Multi-mode chips from TI, Freescale, IDT, etc, support WPC and PMA standards, and are promising A4WP support as well. Intel apparently favours A4WP because it extends the capabilities of wireless charging beyond mobile phones to laptops and ultra-books too.

According to Kumar from Texas Instruments, “TI’s wireless power transmitter and receiver products are used in more than 80 per cent of wireless power products released in the market. For example, currently all Nokia Lumia phones and chargers supporting wireless power are designed using TI products. Our portfolio of transceivers include innovative technologies, a broad range for all applications, comprehensive design support and reference designs, app notes, and complete solutions. It helps design engineers to develop innovative, efficient wireless charging capabilities

for smartphones, tablets and other portable electronics, and design wireless power charging sources, ranging from pads to sources used in cars and furniture. Moreover, TI is an industry leader with the first WPC 1.1 ICs.”

Asked about the merits of specific products, he replies, “Specifically, TI’s bq5101x integrated receivers provide a regulated DC output and digital-control feedback to the transmitter, while TI’s bq500xx family of wireless power transmitters efficiently manages the power transfer to the receiver.”

TI also offers development aids, and support to engineers working in this emerging field. Whether implementing wireless charging within an existing design or adding this functionality to a new one, engineers can make their own kit by selecting a transmitter/receiver evaluation module. TI has evaluation modules for bq500410AEVM-085, bq500211EVM-045 and bq51013EVM-725. Along with development tools, TI also provides a large selection of support collateral including technical documentation, reference designs, application expertise, customer support, and third-party and university programmes. “Our design experts on the E2E community give engineers an interactive platform to get their questions addressed. Also, as mentioned, the extensive network of third-parties consists of recommended companies, RF consultants, and independent design houses that provide a series of hardware module products and design services,” adds Kumar.

Most component makers and solution providers are betting big on the wireless charging market, and trying to include all leading technologies in their product range. “We predict rapid growth of wireless charging technology in many applications over the next five years. We’d like to expand the market shares of our coils by manufacturing solutions for all major wireless power standards, including WPC, A4WP and PMA in the near future,” says Yasunori Terajima, head of new business promotion group of the electronic components sales & marketing group at TDK Cor-

poration, a leading provider of wireless power transfer coil units for use in smart phones and other mobile devices.

Who will win the race?

It is evident that there is a lot of work happening in this space and a lot of competition too. Apart from the standards race, there is a lot of enthusiasm in developing new products too. There are several mind-boggling technologies, including Eurobalise (a wireless charging system between trains), Duracell Powermat’s WiCC (an after-market memory card-like insert with a next-generation wireless power receiver, coupled with NFC technology), WiMat (a wireless charger for phones that takes standard USB power input) and Qualcomm Halo (for wireless electric vehicle charging).

“According to IHS, about 5 million wireless charging units were shipped in 2012, and they estimate that 100 million wireless power devices could be in use by 2015. We find that the market today is focused primarily on products for mobile phones, but is expected to include products for industrial, automotive and other consumer goods in the future,” says Kumar, adding, “One significant development that will impact the adoption of wireless charging technology is the technical standards. Currently, there are three standards consortiums—WPC, PMA and A4WP. Consensus on a universal standard is important to the development/evolution of this technology as it will drive wider adoption of wireless charging, and ensure customer satisfaction and backward-compatibility as the technology evolves.”

We will have to wait and see who wins this race. Daniel Schreiber, president of Powermat, stated in an interview with ‘The Verge’ that standards are ultimately set in a coffee shop and not in a conference room. He reminded readers of the case of Wi-Fi versus HomeRF, where Starbucks’ endorsement of Wi-Fi in 2001 gave the technology irreversible momentum. So, will Starbucks’ backing do the magic for PMA too? ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai.